

Supervised Machine Learning Hand Gesture Classification in VR for Immersive Training

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ABSTRACT

The fast adoption of immersive wearables evidences the need for more intuitive interaction methods between virtual environments and their users. Advances in wearables are making in-built real-time hand tracking mechanisms more common. However, wearable providers only include limited gestures available for developer use. This limits their use for VR-based training solutions, where users need to complete tasks that mimic real-world activities as much as possible to gain valuable insights on those tasks. We present a virtual reality application that collects data on hand characteristics and analyses the collected data to identify hand gestures towards achieving a more natural interaction.

Keywords: Virtual reality, machine learning, classification.

Index Terms: Human-centered computing—Human-computer Interaction (HCI)—Interaction paradigms — Virtual Reality. Human-centered computing—Human-computer Interaction (HCI)—Interaction techniques — Gestural input. Computing methodologies—Machine learning—Learning paradigms—Supervised learning.

1 INTRODUCTION

The increasing complexity of computer interfaces demands innovative interactions with its users. Hands are humans' natural everyday tools to interact with physical objects. Advancements in real-time tracking technology are making in-built hand tracking more common in augmented reality (AR), virtual reality (VR) and mixed reality (MR) wearables. Organisations are using immersive training to transfer and maintain knowledge within their workforce. However, one of the challenges is to find an interactive wearable that is easy to use for a diverse group of people. Wearable controllers vary in shape, size, ergonomics, and feedback. This makes it difficult for users to get accustomed to different wearables, negatively impacting users with less controller-based device experience. Specifically, employees without gaming experience struggle with controllers [1]. However, hand tracking is now a viable option to replace hand-held controllers. Immersive wearable development kits and vendors provide the foundations for developers to program hand interactions; yet, they tend to work for specific devices resulting in low portability. This results in more work for the developers to refactor their application to work in different wearables.

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2 LITERATURE REVIEW

The hand gesture recognition process involves collecting data on hand characteristics and analysing it to classify the gestures. A static gesture comprises the hand characteristics at a given time (e.g., posture, position). Therefore, the previous and later forms of the hand are irrelevant. A dynamic gesture is a group of static gestures captured in a period. Dynamic gestures can also be referred to as actions [2]. Approaches for capturing hand features include wearable sensors (e.g., haptic gloves) and vision-based analysis (e.g., cameras). In the case of wearable sensors, they use electromyography (EMG) signals to detect changes in muscle movements [3]. In addition, they use flex sensors to capture information on fingers' joint angles and gyroscopes to measure the orientation of hands [4]. Vision-based approaches are used to extract information about users' hand joints [5], velocity and position [6], and hand shape [7]. The advancements in technology and devices' cost reduction have made it possible to introduce high-end vision-based motion detection cameras (e.g., Leap Motion controllers, Azure Kinect, and Intel RealSense Depth Camera), providing better tracking.

2.1 Proposed System

Our work proposes a hand data collection and classification system built using Unity3D's physics engine¹ and the Oculus Integration package to adapt non-traditional gestures for immersive training. Data collection and experiments were done using the Oculus Quest 2 headset. The headset uses eight inside-out cameras to track hands (and controller) movements from the physical world. This paper reports the accuracy of the supervised classification methods and the type of hand data characteristics used.



Figure 1: Fingertip-to-palm-centre distance (left) and rotation and position data points (right).

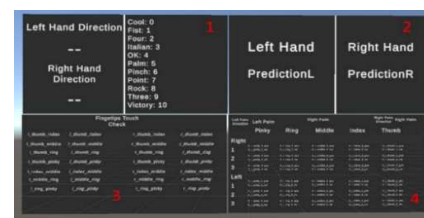


Figure 2: VR application.

2.2 Data Collection

Our application collected data points on each hand every second. In total, we collected 181 rotation and position data points on fingers and palm, 10 fingertip-to-palm-centre distance data points and 14 binary values (0,1) showing when fingertips were in contact with each other. The data was stored locally as a Comma Separated Vector (CSV) file. Figure 1 shows the points where the distance from fingertip to palm centre was taken (left) and the points where the rotation and position data were collected (right). Figure 2 shows the VR application that allows the user to visualise data capture and classification in real-time. Canvases 1 and 2 in Figure 2 showed the predictions from Section 2.4, and Canvases 3 and 4 visualised the prediction's information.

2.3 Experiments

We experimented on 11 different hand gestures; they were “Cool”, “Fist”, “Palm”, “Four”, “Three”, “Italian Appreciation Gesture”, “Thumbs Up”, “Pinch”, “Rock ‘n Roll”, “Victory” and “Pointing”. Data for each gesture was generated using the data collection application described in Section 2.2. This data was then later labelled and used to train four supervised learning methods: K-Nearest Neighbour (KNN), Random Forest (RF), Gradient Boosting (GR) and Support Vector Machine (SVM). We divided the data into two groups. Group A had the data on the fingertip to palm centre distance, and Group B had data on rotation and position of points in hand. Each data group was divided into two subsets of train and test. The training set had 80 percentiles of the data, and testing had the rest. Data Group B suffered from the curse of dimensionality; therefore, Principal Component Analysis (PCA) was used to reduce the dimensions. The top 3 new features created by PCA were used in Group B classification.

Table 1: Accuracy results per classification method.

Method	Group A Accuracy (%)	Group B Accuracy (%)
KNN	99%	86%
RF	99%	87%
GR	92%	81%
SVM	75%	69%

2.4 Dynamic Demonstrator

The KNN machine learning model with Group A dataset was exposed to the VR interface using web services (Fig. 3) via Flask, a Python-based micro web framework. Information on each hand was sent to the classifier with a fixed time interval of one second. Then, the classifier returns the outcome to the VR interface for the user to see it (Fig. 2). In general, the outcome for all gestures was classified correctly. However, sometimes there was a delay in the classification result when the user moved too quickly or if the hands were not in the headset’s field of view (FOV).

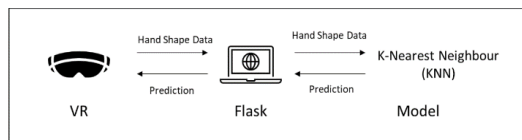


Figure 3: System architecture.

3 DISCUSSION

We used the distance and fingertip data to classify eleven different hand gestures. Some hand gestures showed less accuracy in

classification than others due to similar notions. For example, hand gestures such as the “Italian Appreciation Gesture” and “Pinch” were the least accurately classified. Therefore, different hand characteristics can be included to classify two gestures with higher accuracy. The position and rotation data did not achieve high accuracy in identifying hand gestures; however, this information can be used to simulate the trajectories of the hand in a virtual environment. Furthermore, the experiments did not include any orientation or direction-dependent gestures. The machine learning methods used in Section 2.3 would not classify gestures such as “Palm-Facing User” or “Palm-Not Facing User”.

4 CONCLUSION

The paper investigates gestures classification using the in-built hand tracking mechanism provided by the Oculus Quest 2 headset. To do so, we built a data collection application to look at different characteristics of the hand and implemented supervised machine learning methods for classification. We achieved an optimal accuracy of 99 percentile using KNN and RF with Data Group A. Data Group B had a larger dimension thus required more data or a method to reduce dimension without loss of information. KNN worked well with our data on distance measure and chosen group of 11 gestures. However, the addition of similar gestures would increase the chance of misclassification. Future work includes looking at direction and orientation-dependent gestures and expanding the system to classify actions. This will help develop more natural interaction building blocks in virtual environments and further assist in understanding user behaviour in immersive training activities towards using more natural hand actions to provide a better user experience for the workforce.

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ⁱ Unity3d - <https://docs.unity3d.com/Manual/PhysicsSection.html>